

CHE 435/525 *Process Systems Analysis and Control*

## Lecture 5: **Further Properties of Laplace Transform**

Wentao Tang

Assistant Professor, Chemical & Biomolecular Engineering  
North Carolina State University

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# Discuss & Recapture

- How do the Laplace transforms of step, impulse, exponential, sinusoidal signals look like?



- How to do partial fraction expansion for any given fractional function of  $s$ ?
- How to use Laplace transform to solve ODEs?

# Learning objectives of this lecture

- ① Further properties of Laplace transform: Scaling of  $t$ , multiplication by  $t$ , final and initial value
- ② Use Laplace transform for process examples described by ODEs
- ③ Analyze the qualitative behaviors of a signal from its Laplace transform

# Laplace transform

## Definition

Given a function (“signal”)  $f(t)$  on  $t \geq 0$ , its Laplace transform is

$$F(s) = \mathcal{L}[f(t)] = \int_0^{\infty} f(t)e^{-st} dt \quad (1)$$

comprehended as a function of a **complex variable**  $s$ .

- Property 3: Scaling of  $t$

$$\mathcal{L}[f(at)] = \int_0^{\infty} f(at)e^{-st} dt = \int_0^{\infty} f(\bar{t})e^{-s\bar{t}/a} \frac{1}{a} d\bar{t} = \frac{1}{a} F\left(\frac{s}{a}\right).$$

- Property 4: Multiplication by  $t$

$$\mathcal{L}[tf(t)] = \int_0^{\infty} tf(t)e^{-st} dt = \int_0^{\infty} -\frac{d}{ds} [f(t)e^{-st}] dt = -F'(s).$$

# Practice problems

- Exercise 1:  $\mathcal{L}^{-1}\left[\frac{1}{(s+a)^2}\right] = ?$
- Exercise 2:  $\mathcal{L}^{-1}\left[\frac{1}{\tau s+1}\right] = ?$

# Final and initial value theorems

- **Final value theorem:** If  $\lim_{t \rightarrow \infty} y(t)$  **exists**, then

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s).$$

- ▶ The condition must not be violated! For  $y(t) = e^t$ ,  $Y(s) = 1/(s - 1)$ ,

$$\infty = \lim_{t \rightarrow \infty} y(t) \neq \lim_{s \rightarrow 0} sY(s) = 0.$$

- **Initial value theorem:**

$$\lim_{t \rightarrow 0} y(t) = \lim_{s \rightarrow \infty} sY(s).$$

# SISO system modeled as ODE

- The reason for doing L.T. is to deal with **process dynamics**, described by linear or nonlinear ODEs.
- If we have only one input (MV or DV) and only one output, the system is called **single-input-single-output (SISO)**.
- In general, a **linear** SISO system can be expressed as

$$\begin{aligned} \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = \\ \frac{d^m u}{dt^m} + b_{m-1} \frac{d^{m-1} u}{dt^{m-1}} + \cdots + b_1 \frac{du}{dt} + b_0 u \end{aligned} \quad (2)$$

where  $y(t)$  is the output and  $u(t)$  is the input.

# SISO system modeled as transfer function

- We are interested in how  $y(t)$  respond to  $u(t)$ . Hence, we usually assume initial conditions  $y(0) = 0$ ,  $y'(0) = 0$ , ...,  $y^{(n-1)}(0) = 0$ .
- Use Laplace transform

$$s^n Y(s) + a_{n-1} s^{n-1} Y(s) + \cdots + a_1 s Y(s) + a_0 Y(s) = b_m s^m U(s) + b_{m-1} s^{m-1} U(s) + \cdots + b_1 s U(s) + b_0 U(s).$$

- We obtain

$$Y(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} U(s).$$

## Definition

The **factor**  $Y(s)/U(s) = G(s)$  is called the **transfer function** from the input to the output

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}.$$



# Calculating output response via Laplace transform

## Seborg Ex.3.4(a)

Given the process model

$$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 3u(t),$$

where  $y(0) = 0$ ,  $y'(t) = 0$ , find  $y(t)$  for  $u(t) = be^{-2t}$ .

① By L.T.,  $(s^2 + 6s + 8)Y(s) = 3U(s)$ , i.e.,  $G(s) = 3/(s^2 + 6s + 8)$ .

②  $U(s) = b/(s + 2)$ . Thus,

$$Y(s) = \frac{3b}{(s + 2)(s^2 + 6s + 8)} = \frac{3b}{(s + 2)^2(s + 4)}.$$

③ Do partial fraction expansion:

$$Y(s) = \frac{3b}{(s + 2)^2(s + 4)} = \frac{a_1}{s + 2} + \frac{a_2}{(s + 2)^2} + \frac{a_3}{s + 4},$$

# Calculating output response by Laplace transform

## Seborg Ex.3.4(a)

Given the process model

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- ⑥ Determine coefficients:

$$3b = a_1(s+2)(s+4) + a_2(s+4) + a_3(s+2)^2$$

by  $3b = 2a_2$  (by setting  $s = -2$ ),  $3b = 4a_3$  (by setting  $s = -4$ ), and  $a_1 + a_3 = 0$ .

- ⑦ Thus,  $Y(s) = \frac{-3b/4}{s+2} + \frac{3b/2}{(s+2)^2} + \frac{3b/4}{s+4}$ .

$$y(t) = \mathcal{L}^{-1}[Y(s)] = -\frac{3b}{4}e^{-2t} + \frac{3b}{2}te^{-2t} + \frac{3b}{4}e^{-4t}.$$

# Key observation

- The **values and multiplicities** of the **roots of denominator** determine what terms will appear after inverse L.T.

$$Y(s) = \frac{\cdots}{\cdots (s-r)^k \cdots} \Rightarrow Y(s) = \cdots + \frac{\cdots}{(s-r)^k} + \cdots$$
$$\Rightarrow y(t) = \cdots + \text{const} \cdot t^{k-1} e^{rt} + \cdots$$

- When  $r = a + bi$ ,  $t^{k-1} e^{rt} = t^{k-1} e^{at} (\cos bt + i \sin bt)$ .
  - ▶ If  $a = \text{Re } r < 0$  ( $r$  is on the **left half plane**), then the term **decays exponentially**.
  - ▶ If  $a = \text{Re } r > 0$  ( $r$  is on the **right half plane**), then the term **grows exponentially**.
  - ▶ If  $a = \text{Re } r = 0$  and  $k \geq 2$  (**repeated root on the imaginary axis**), then the term **grows polynomially** ( $\propto t^{k-1}$ ).
  - ▶ Whenever  $b = \text{Im } r \neq 0$  ( $r$  is not real), the term is **oscillatory**.
  - ▶ If  $a = \text{Re } r = 0$ ,  $b = \text{Im } r \neq 0$  and  $k = 1$  (**simple imaginary root**), then the term **persistently oscillates**.
  - ▶ If  $a = \text{Re } r = 0$ ,  $b = \text{Im } r = 0$  and  $k = 1$  (**simple root at 0**), then the term is **constant**.

# Qualitative behavior analysis

- Thus, given  $Y(s)$ , we can know how  $y(t)$  behaves as  $t$  increases.
  - ▶  $y(t)$  is dominated by the term(s)  $t^{k-1}e^{rt}$  whose  $r$  has the right-most real part.
  - ▶ Think about

$$y(t) = -2e^{(1+3i)t} + 5e^{2it} - 4 - 7e^{-2-10i}$$

When  $t$  is large enough, which term dominates?

# Qualitative behavior analysis

## Practice: Seborg Ex.3.9

Determine **what functions of time** will appear in  $y(t)$  in the following cases. Which  $y(t)$  are **oscillatory**? Which have a **constant value** for large values of  $t$ ?

①  $Y(s) = \frac{2}{s^2(s+4)}$

②  $Y(s) = \frac{2}{s(s+1)(s+3)}$

③  $Y(s) = \frac{2}{s(s+2)^2}$

④  $Y(s) = \frac{2}{s(s^2+4s+8)}$

⑤  $Y(s) = \frac{2(s+1)}{s(s^2+4)}$